AIR RENTAL

Developer Document

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1. Product Overview

- All platforms use the same backend, API, database, authentication, permissions and business rules.
- Country-specific utilities, payments, currencies and local rules must be added through modular integrations.
- Pakistan has first priority for local utility and payment integrations.
- V1 revenue model: subscription only. Marketplace and transaction-based revenue are future options.

2. Onboarding & Authentication

- Show language selection first, with search.
- Public role options: Landlord / Owner, Tenant / Lessee and Manager / Agent.
- Landlord / Owner and Tenant / Lessee must choose an Account Category: Individual or Business / Organization.
- Individual: the account is managed in the person's name and the dashboard/profile shows the personal name.
- Business / Organization: for companies, shops, agencies, schools, NGOs, trusts and other organizations; show the business/organization name and logo when available.
- Account Category is only a category. It does not create another portal.
- One verified email address can belong to only one Air Rental account.
- Allow sign-up/sign-in with Google, Apple or Manual Sign-Up. Use OTP verification where required.
- If Google, Apple or Manual Sign-Up matches an existing verified email, use or securely link the existing account instead of creating a duplicate.
- After setup, allow Face ID / Touch ID login or app unlock on supported devices.
- One account may have multiple public roles. Show only active roles in the role switcher and allow switching without logout.
- Internal Admin is never shown in public signup, role selection or role switching.
- Allow language changes later in Settings.

3. Global Country, City & Language System

- Property country is selected manually. Do not auto-select it from GPS or device location.
- Provide searchable country and city/region lists.
- City/region options should follow the selected property country.
- Store country, currency, measurement units and address settings at property level.
- One account can manage properties in multiple countries.
- Supported languages: English, Chinese (Mandarin), Hindi, Spanish, Arabic, French, Bengali, Portuguese, Russian, Urdu and Roman Urdu.

4. Property Registration & IDs

- Property types: House, Apartment, Shop, Floor, Room, Office, Warehouse and Plot.
- Visible type codes: HSE, APT, SHP, FLR, ROM, OFC, WRH and PLT.
- Generate simple visible IDs such as APT-1 and APT-2.
- Also store a hidden permanent global UUID for every property record.
- The same address may contain multiple units/properties.
- Use relevant size units: Marla for houses in Pakistan, square feet where appropriate, and local units in other markets.
- If ownership changes, keep the property code and UUID with the same property record; update ownership and access only.

5. Property Profile & Dashboard

- Store property name, code, type, country, city, area, full address, postal/ZIP code, map location, size, facilities, photos and documents.
- Property status: Rented or Vacant.
- Landlord / Owner dashboard shows all owned or authorized properties.
- For Individual accounts, show the personal name. For Business / Organization accounts, show the business/organization name and logo when available.
- Landlord / Owner property tools: Tenant / Lessee, Manager Assignments, Rent, Bills, Repairs, Hisab, Inspection, Documents, Chat, Alerts, AI and Reports.
- Tenant / Lessee dashboard shows only rented properties/locations and related Rent, Bills, Payments, Agreements, Maintenance, Documents, Chat and Notifications.
- Manager / Agent dashboard shows only assigned properties and permitted features, including Chat when allowed.
- Property lists should support search and filters for Country, City, Property Type, Rented/Vacant and Rent Due/Overdue.

6. Tenant / Lessee Account & Limited Access

- Tenant / Lessee can access only assigned tenancy/property data.
- Allow invitation by selected contact, secure link or QR code.
- Invitation links should be secure, time-limited and single-use where practical.

- Existing Air Rental user: show the invitation as an in-app alert and push notification when enabled. Accepting links the tenancy to the same account.
- New user: open the secure signup flow, verify/create the account, then show the pending invitation for Accept or Decline.
- Tenant / Lessee may assign a Manager / Agent to specific rented properties with limited permissions.
- Each active tenancy has one private shared chat for the linked Landlord / Owner, Tenant / Lessee and authorized Manager / Agent.
- Chat participants can send and read messages and share pictures, videos and documents. Show timestamps, read receipts and new-message notifications.
- Keep chats separate by property and tenancy. A new tenancy starts a new chat, and previous tenancy chats remain separate. Only authorized participants can access each chat.
- Tenant / Lessee cannot see the owner's other properties, unrelated tenants or private owner finances.
- Business / Organization tenants may manage multiple rented locations in the same Tenant / Lessee Portal.

7. Tenant / Lessee Profile - Identity

- Basic profile: full name, mobile number, email and current address.
- For Pakistan, allow identity selection and upload for CNIC, National Identity Card or Driving License, including front/back where applicable.
- Use equivalent identity documents in other countries.
- Emergency contact is optional.
- Identity documents are sensitive and must be encrypted, securely stored and permission-restricted.

8. Tenant / Lessee Rent Agreement / Tenancy Contract

- Link every agreement to the property and the Tenant / Lessee account.
- Allow agreement upload as PDF or image.
- Store start date, expiry date, monthly rent, security deposit, rent due date, renewal terms and other terms.
- Landlord / Owner manages agreements for owned/authorized properties.
- Manager / Agent may manage an agreement only when the assignment includes that permission.
- Tenant / Lessee can view only its own agreement.
- Keep previous agreements in history after renewal.

9. Contract Renewal Notification

- Monitor agreement expiry automatically.
- During the final two months, remind the Landlord / Owner on the 1st day of each month.
- Renewal options: 1 to 30 years or Custom End Date.
- On renewal, calculate the new expiry date, save the renewal record, keep the old agreement, attach the new agreement and notify the Tenant / Lessee and relevant Manager / Agent.
- Keep a complete renewal timeline.

10. Manager / Agent System & Permissions

- Manager / Agent Portal is separate from the Landlord / Owner Portal.
- A Manager / Agent may be invited by contact, secure link or QR code.
- A Landlord / Owner or Tenant / Lessee, whether Individual or Business / Organization, may assign a Manager / Agent to specific properties.
- Each assignment must store the assignment side (Owner side or Tenant side), properties and granted permissions.
- Manager / Agent can see only assigned properties and allowed features.
- Permission options may include Tenant / Lessee details, rent, bills, repairs, inspections, documents, chat and tasks.
- A Manager / Agent cannot receive more access than the assigning user has.
- A Tenant-side Manager / Agent cannot access owner-only records or private owner finances.
- Log important Manager / Agent actions for the assigning user and authorized Internal Admin users.

11. Rent Management

- Landlord / Owner sets monthly rent, due date, currency and terms.
- An authorized Owner-side Manager / Agent may manage rent when permission is granted.
- From any rented property, Landlord / Owner can send the linked Tenant / Lessee a one-click rent reminder.
- Also support automatic rent reminders and optional manual reminders.
- Tenant / Lessee can view rent obligations, due dates and payment history and upload payment proof.
- Statuses: Upcoming, Due, Partial, Proof Submitted, Under Review, Paid and Overdue.
- For partial payment, show Total Due, Paid Amount and Remaining Amount.
- Payment proof remains pending until approved by the Landlord / Owner or an authorized Owner-side Manager / Agent.
- AI may extract amount, month/date, payment method and reference number from uploaded proof.
- Rent Statement should combine rent history and receipts in one view.

- Do not show Due Date and Paid Date as separate statement-list columns. Show them inside the opened transaction details.
- Transaction details should show due date, paid date, payment reference, receipt/proof and approval timeline.
- Allow custom date-range statement export in PDF, Excel and CSV.

12. Bills - Responsibility & Notifications

- Bill responsibility options: Landlord / Owner, Tenant / Lessee, Manager / Agent or Shared.
- Rented property with Tenant / Lessee responsibility: alert the Tenant / Lessee; after payment, update the Landlord / Owner and relevant Manager / Agent.
- Vacant property: alert the Landlord / Owner and relevant Manager / Agent.
- Rented property with Owner responsibility: alert the Landlord / Owner and relevant Manager / Agent.
- Shared responsibility: notify only configured responsible users.
- Owner-level taxes/challans normally remain on the owner side. Utility responsibility follows the tenancy agreement.

13. Automatic Utility Bill Fetching - Pakistan Priority

- Fetch and update utility bills automatically where official or authorized access exists.
- Pakistan priority providers may include SNGPL, SSGC, IESCO, LESCO, K-Electric, MEPCO, FESCO, GEPCO, PESCO, HESCO, SEPCO and major WASA/municipal providers.
- Verify each provider separately. Do not use unauthorized scraping or bypass methods.
- If direct integration is unavailable, allow bill photo/PDF upload and AI extraction.
- Bill data should include provider, billing period, amount, due date, consumer/reference number and status.

14. Bill Payment & Cards

- Where supported, Landlord / Owner, Tenant / Lessee or an authorized Manager / Agent can pay bills according to bill responsibility and granted permissions.
- Landlord / Owner manages bill records, responsibility and payment status. Store the payer for every payment and show Paid by Landlord / Owner, Paid by Tenant / Lessee or Paid by Manager / Agent.
- Manager / Agent may manage or pay assigned bill records only when permitted. Show payment confirmation and history to authorized users.
- Use a secure payment gateway and tokenized payment methods. Never store full card numbers or CVV on Air Rental servers.
- Basic flow: New Bill, Alert, Bill Details, Pay Bill, Confirmation, Paid Status.
- Bill fetching and bill payment are separate integrations.

15. Maintenance / Repairs

- Tenant / Lessee can report an issue using text, photo, video or optional voice.
- Landlord / Owner reviews repair requests and controls repair decisions.
- Authorized Manager / Agent may review, assign and track repairs for assigned properties.
- Workflow: New, Approved, Assigned, In Progress, Completed.
- Track contractor, estimate, approved amount, final amount, invoice and payment proof.
- Support configurable owner approval thresholds and before/after evidence.

16. Property Inspection

- Landlord / Owner can start an inspection.
- Authorized Manager / Agent can inspect assigned properties when permitted.
- Record photos, electricity/gas/water meter readings, damage, furniture/items and general condition.
- AI may create a short inspection summary.
- Keep dated inspection history.

17. Move-In / Move-Out Handover

- Move-in record: photos, meter readings, keys, existing damage, furniture/items and condition.
- Move-out record: final photos/meters, final rent/bills, damage and condition.
- Track security deposit settlement: Deposit Received, Deductions, Refund and Remaining Balance.
- Record confirmations from Landlord / Owner, Tenant / Lessee and Manager / Agent according to permissions.
- AI may create a short handover report and before/after comparison.

18. Documents & Expiry Alerts

- Store rent agreements, ownership documents, property tax/challans, NOCs, society documents, utility documents and other relevant files.
- Allow PDF and image uploads.

- Insurance documents are optional by default.
- Landlord / Owner can upload Property Insurance or Landlord Insurance for each property.
- Tenant / Lessee can upload Tenant Insurance or Renter Insurance for the relevant tenancy.
- Landlord / Owner can set Tenant Insurance Required ON/OFF for each tenancy or property. If ON, the Tenant / Lessee must upload a valid insurance document.
- Insurance details: Insurance Provider, Policy Number, Start Date, Expiry Date and Document Upload.
- Authorized Manager / Agent can upload and manage insurance documents for a linked Landlord / Owner or Tenant / Lessee only within granted management permissions.
- Landlord / Owner can view the Tenant / Lessee insurance document for the relevant tenancy where access is allowed.
- Tenant / Lessee can see only insurance documents related to their own tenancy. Manager / Agent can see only authorized clients, properties and tenancies.
- Send an expiry reminder before an insurance document or other tracked document expires.
- AI may extract document type, dates, amounts, references, policy details and expiry dates when visible.

19. Property-wise Hisab

- Provide a simple property financial summary, not a full accounting system.
- Track rent income and property expenses for Landlord / Owner.
- Show Net Income with monthly and yearly history.
- Owner-side Manager / Agent sees only permitted owner-side financial items.
- Tenant-side Manager / Agent sees only permitted tenant-side rented-property records and never private owner finances.

20. AI Property Assistant

- Landlord / Owner can ask about owned/authorized properties, rent, bills and permitted financial data.
- Tenant / Lessee can ask only about rented properties and related records.
- Manager / Agent can ask only about assigned properties and data allowed by the assignment.
- AI must follow the same access permissions as the logged-in user.
- AI may summarize receipts, bills, documents, inspections and property photos.
- Keep answers short, practical and in the selected language.
- Important financial or document actions still require user confirmation.

21. Smart Alerts & Notifications

- Important alerts: overdue rent, rent reminders, new/due bills, payment updates, new chat messages, expiring agreements, repairs, inspections and expiring documents.
- Tenant / Lessee receives only alerts relevant to rented properties and its responsibilities.
- Landlord / Owner receives owner-side property, rent, bill, agreement, repair, inspection and document alerts.
- Manager / Agent receives only assigned-task and permitted property alerts.
- Internal Admin receives platform incidents, integration problems, support events and configuration alerts.
- Avoid unnecessary duplicate notifications.

22. Activity / Audit Trail

- Record user, role, portal, property, date/time, action and reference.
- Examples: payment proof submitted, payment approved, bill uploaded, inspection completed and manager assignment changed.
- Ordinary users cannot edit or delete audit events.
- Authorized Internal Admin users can review audit events.

23. Security, Privacy & Data Access

- Enforce role, portal, property-assignment and permission checks on every protected action.
- Landlord / Owner can access only owned or authorized properties.
- Tenant / Lessee can access only assigned rented properties.
- Manager / Agent access is limited to assigned properties, assignment side and granted permissions.
- Internal Admin access is separate, private and restricted.
- Encrypt sensitive identity documents and restrict file access.
- Use secure authentication, sessions and storage.
- Use consent-based contact access and store only required contact data.
- Use secure payment-provider tokenization for payment data.
- AI must use the same permissions as the current user.

24. Recommended Core Backend Entities

- User, Role, UserRole/RoleAssignment, OwnerProfile, TenantProfile and ManagerProfile.
- Organization, OrganizationUser, Property, PropertyAddress, Tenancy and OrganizationPropertyLink.
- ManagerAssignment with assigner, assignment side, property and granted permissions.
- RentSchedule, RentRecord, RentReceipt, TenancyAgreement and AgreementRenewal.
- BillProvider, BillAccount, Bill, BillPayment and PaymentMethodToken.
- MaintenanceRequest, Inspection, Handover, Document, ChatThread, ChatMessage, ChatAttachment, MessageReadReceipt, Notification, ActivityLog, AIExtraction, Subscription and Admin configuration records.
- OwnerProfile and TenantProfile must store Account Category: Individual or Business / Organization. Business / Organization profiles also store name, logo when available and basic contact details.
- Every property has a permanent internal UUID and a simple visible property code.
- One User can hold multiple public roles without duplicate accounts.

25. Recommended Screen Map

- Onboarding: Language Search, Public Role, Account Category for Landlord / Owner or Tenant / Lessee, then Sign Up or Login.
- Landlord / Owner Portal: Home, Properties, Tenant / Lessee, Manager Assignments, Rent, Bills, Repairs, Inspection, Documents, Chat, Hisab, Reports, Notifications and AI.
- Tenant / Lessee Portal: Home, My Rented Properties/Locations, Manager Assignments, Rent, Bills, Payments, Agreements, Maintenance, Documents, Chat, Notifications and AI.
- Manager / Agent Portal: Home, Assigned Properties, Tasks, Tenant / Lessee, Rent, Bills, Repairs, Inspection, Documents, Chat, Reports, Notifications and AI. Screens/actions depend on each assignment's side and permissions.
- Internal Admin Portal: private Users, Organizations, Providers, Subscriptions, Support, Audit, Integrations, Settings and Analytics. Final access method and detailed permissions will be agreed with the developer.
- Shared public screens: Profile, Settings, Language, Subscription/Billing, Help, Privacy and Security.
- Public role switching uses the same login and never includes Internal Admin.

26. MVP Must-Have Features

- Google, Apple and Manual authentication; one verified email per account; OTP where required; biometric login/app unlock.
- Four portal foundations: Landlord / Owner, Tenant / Lessee, Manager / Agent and private Internal Admin.
- Global country/city system, property registration, permanent IDs, search and basic filters.
- Landlord / Owner and Tenant / Lessee account categories: Individual or Business / Organization, including business/organization name and optional logo.
- Tenant / Lessee invitation, identity upload, agreement and renewal.
- Manager / Agent assignment by Landlord / Owner or Tenant / Lessee with granular permissions.
- Private tenancy chat for Landlord / Owner, Tenant / Lessee and authorized Manager / Agent, with messages, file sharing, notifications, timestamps and read receipts.
- Single Account / Multiple Roles with role switching.
- Rent tracking, partial payments, reminders, payment-proof approval, statement/history and export.
- Bill responsibility, Pakistan utility integration framework, bill upload/AI fallback and secure payment layer where supported.
- Maintenance, inspections, move-in/move-out and security deposit settlement.
- Documents/expiry alerts, Hisab, notifications, audit trail and basic AI assistant.
- Business / Organization Tenant support for multiple rented locations.
- Subscription plans use total active unit count to select one plan, and that plan's per-unit rate applies to all active units. All plans have the same features: Starter (1-5 units) $10/unit/month; Standard (6-15) $8/unit/month; Advanced (16-50) $6/unit/month; Professional (51-150) $4/unit/month; Agency (151+) $2/unit/month. Starter includes one 30-day free trial for 1 unit per account.

27. Future Features

- Marketplace.
- More online rent collection and utility/payment integrations.
- Advanced vendor management.
- Digital agreements and e-signatures.
- Tenant screening and bank aggregation.
- Advanced AI rent recommendations.
- Additional countries and provider integrations.
- Billing APIs for additional countries.

28. UI/UX Principles

- Keep the app simple for ordinary and less-technical users.
- Use large buttons, clear icons, short labels, search bars and clear statuses.
- Show only fields and actions relevant to the selected public role and current permissions.
- Keep Internal Admin completely separate from public navigation.
- Use consistent navigation, navy headers, clean cards and clear primary action buttons.
- Keep rent/payment lists simple; show full details only after opening a transaction.
- Use clear statuses such as Paid, Partial, Due, Overdue, Pending and Completed.
- Use short, simple wording in all supported languages.
- Do not overload dashboards.

29. Critical Business Rules

- Account Category does not create a separate portal.
- One public account may hold Landlord / Owner, Tenant / Lessee and Manager / Agent roles at the same time.
- Each role context shows only its own permitted properties and data.
- Manager / Agent permissions cannot exceed the assigning Landlord / Owner or Tenant / Lessee user's own rights.
- Internal Admin is private and separate from all public roles.
- Financial status changes based on user-submitted proof require authorized approval where defined.
- Use only official or authorized external integrations.

30. High-Level Acceptance Criteria

- Google, Apple and Manual Sign-Up work; duplicate verified email does not create a second account.
- Face ID / Touch ID works on supported devices after account setup.
- A user with multiple public roles can switch roles with one login and sees only active roles.
- Landlord / Owner and Tenant / Lessee can select Individual or Business / Organization; the correct personal or organization name/logo appears.
- Landlord / Owner can add properties and see all owned/authorized properties.
- Tenant invitation works for existing and new users without creating duplicate accounts.
- Manager / Agent can be assigned by either Landlord / Owner or Tenant / Lessee and sees only assigned properties and permissions.
- Shared tenancy chat works only for authorized participants, supports pictures, videos and documents, shows timestamps and read receipts, sends new-message notifications, and keeps each property/tenancy chat separate.
- One-click and automatic rent reminders work.
- Partial rent, payment-proof approval and rent-statement export work correctly.
- Renewal reminders and 1-30 year/custom renewal options work and preserve agreement history.
- Bill responsibility, authorized auto-fetch or upload/AI fallback, payment status and Paid by payer attribution work correctly.
- Maintenance, inspections, handover and deposit settlement can be recorded.
- Insurance upload, Tenant Insurance Required ON/OFF, expiry reminders and role-based insurance access work correctly.
- Subscription billing selects the plan from total active units, applies that plan's per-unit rate to all active units, keeps the same features across plans, and gives Starter one 30-day free trial for 1 unit per account.
- AI never returns data outside the current user's role, property assignment or permission scope.

31. Developer Implementation Notes

- Keep architecture modular for countries, languages, currencies, property types, utility providers and payment providers.
- Mobile and web must use the same backend/API, permissions and business rules.
- Use official/authorized utility integrations only.
- Use a secure payment gateway; never store full card number or CVV.
- Base notification logic on role, portal, property status and responsibility.
- Use background jobs, webhooks or permitted polling only where needed.
- Prevent duplicate bill and payment records.
- AI extraction should return structured fields and indicate when user verification is needed.
- Keep Internal Admin controls separate from public signup, role selection, role switching, normal navigation and public website menus.
- Final Internal Admin access method and detailed permissions will be agreed with the developer.
- Allow one User record to hold multiple public roles.
- Use the phone's secure biometric system for Face ID / Touch ID. Air Rental must not store raw biometric data.

Core Vision

Air Rental's vision is to make property management simple for individuals, businesses and organizations through one secure and scalable platform. It will support Landlord / Owner, Tenant / Lessee and Manager / Agent portals, with a private Internal Admin Portal for platform administration. The product will be available on iOS, Android and responsive web, initially launching in Pakistan, USA and the UK, with a global-ready foundation for future worldwide expansion. It will keep permissions, payments, documents and property data secure and organized.